University Number: _____

THE UNIVERSITY OF HONG KONG

FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE

COMP7104/DASC7104 - Advanced Database Systems

Date: December 13, 2023 **Time: 6:30pm-8:30pm**

<INSTRUCTION ABOUT NUMBER OF QUESTION TO BE ANSWERED: OVERALL AND FROM EACH SECTION>
23 questions overall, divided into two sections as follows:
Section 1 – Multiple-choice questions – 16 questions
Section 2 – Detailed-answer questions – 8 questions

<INDICATION OF COMPULSORY QUESTIONS>
None.

<INFORMATION ABOUT THE MARK VALUE OF QUESTIONS>
Section 1 – 1.5 points per question – 24/100 points in total
Section 2 – number of points depends on the question – 76/100 points in total

In Section 2, write your answers in the rectangle boxes.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Candidates are permitted to bring one sheet of A4-sized paper with printed or written notes on both sides to the examination.

Mark your University Number on each page (top right corner).

University Number: _____

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General indications

- For questions involving arithmetic expressions, numerical work or derivations, credit will be given only if the work leading to the answer is shown.
- You can leave arithmetic expressions unsolved, as long they are correct and complete this will be considered a valid answer. Example : (M+N) is not a correct and complete answer, (1000+200) is correct and complete if M=1000, N=200, and the formula to apply is M+N.
- 1 KB = 1024 bytes; i.e., we will be using powers of 2, not powers of 10

Section 1 – Multiple-choice questions – 16 questions – 1.5 point per question

Circle the true answer(s) to each question. Each question may have zero, one, or several valid answers. For each question, only the identification of all and only the valid answers will allow you to obtain the assigned point.

Q1.1 Which of the following assertions about join algorithms are true?

(A) In a page nested loop join (PNLJ) where both tables fit entirely in memory, the choice of which table to use as the inner/outer table may affects I/O cost.

(B) In a page nested loop join (PNLJ) where one of the tables fits entirely in memory, it is better to use that table as the inner one.

(C) In an index nested loop join (INLJ) where only one of the tables has an index on the join attribute, the choice of inner/outer branch should be made based on the table sizes.

Q1.2 Which of the following assertions about the System R / Selinger algorithm are true?

(A) It never considers plans with cartesian products because they can always be expressed as a join.

(B) It only explores right-deep plans.

Q1.3 Which of the following assertions about left-deep plans are true?

(A) Two left-deep plans can differ in the order of relations and produce the same output.

(B) Two left-deep plans can differ in the access method for each leaf operator and produce the same output.

(C) Two left-deep plans can differ in the join method for each join operator and produce the same output.

Explanation: all true as discussed on class.

Q1.4 Which of the following is NOT a good rule for rewriting and optimizing relational algebra plans ?

(A) Push projection below the right (inner) branch of a CNLJ (Chunk Nested Loop Join).

(B) Push selection below the right (inner) branch of a CNLJ (Chunk Nested Loop Join).

(C) Push selection below the left (outer) branch of a CNLJ (Chunk Nested Loop Join).

Q1.5 Which of the following join operators is a pipeline breaker?

(A) Index Nested Loop Join.

(B) Sort Merge Join.

(C) Grace Hash Join.

Q1.6 Which assumptions are followed in the selectivity estimator of the System R / Selinger algorithm?

(A) Independence of predicates.

(B) Uniform distribution within histogram bins (when histograms available).

(C) Normal distribution for integer attributes.

Q1.7 Which of the following assertions are true about transaction management in RDBMSs?

(A) All serial schedules are also serializable.

(B) Interleaving transactions is always faster than doing them one at a time.

(C) All conflict-serializable schedules are also recoverable.

Q1.8 Which of the following properties are true about the 2PL / strict 2PL protocol?

(A) strict 2PL enables less schedules than 2PL.

(B) strict 2PL requires that transactions acquire locks gradually, and release them all all at once at the end of the transaction.

(C) strict 2PL ensures that all schedules are recoverable.

Q1.9 Which of the following properties are true about the 2PL / strict 2PL protocol?

(A) Some conflict serializable schedules cannot be produced when using 2PL.

(B) Schedules that are conflict-serializable will not produce a cyclic dependency graph.

(C) Both strict 2PL and 2PL enforce conflict serializability.

Q1.10 Which of the following assertions are true about data partitioning and query evaluation in PDBMSs?

(A) If we partition data using the round robin scheme, look-up by key will incur a higher cost than with range or hash partitioning.

(B) If we partition data using the range scheme, look-up by key will incur a higher cost than using the hash scheme.

(C) A join over two tables that requires a symmetric shuffle will incur fewer network I/Os than one that requires an asymmetric shuffle.

Q1.11 We know that 50% of the records in table Person in our database all share the same value for the attribute *name*. Assume we run a series of experiments, each time increasing the number of machines over which we hash-partition the Person table *by name*, and measure the time to complete a parallel scan

of this table each time. Across experiments, the completion time for parallel scan will be a function of the number of machines used.

(A) Constant.

(B) Exponential.

(C) Linear.

Q1.12 Which of the following assertions are true about data partitioning and query evaluation in PDBMSs?

(A) If data is partitioned using round robin, when searching for a record, we have to perform broadcast lookup to all the machines.

(B) Range partitioning is the only partitioning scheme that ensures balanced load, i.e., each node gets a similar amount of data.

(C) Round-robin partitioning is the only partitioning scheme that ensures balanced load, i.e., each node gets a similar amount of data.

Q1.13 Which of the following assertions are true about PDBMS query evaluation?

(A) Pipeline parallelism scales up with the amount of data.

(B) Partition parallelism scales up with the amount of data.

(C) Pipeline parallelism scales up to the pipeline depth.

Q1.14 Which of the following assertions are true about PDBMS query evaluation?

(A) There is no extra disk I/O cost for parallel Grace Hash Join when we re-partitioning (shuffling) the data across machines.

(B) Data skew may be a problem regardless the partitioning scheme.

(C) Parallel aggregates require that we compute the same aggregation function locally then globally.

Q1.15 Which of the following are valid reasons to replicate data in NoSQL database ?

(A) To handle ever-increasing database sizes.

(B) To recover from failures.

(C) To enable workload scalability.

Q1.16 Which of the following assertions are true for NoSQL database systems ?

(A) They were designed to cope with high rates of simple read/write operations on relational data.

(B) They were designed to cope with long running read-only analytical queries.

(C) They were designed for mixed analytical / transactional workloads.

Section 2 – Detailed-answer questions – 8 questions – 76 points in total

Q2.1 – **Join Algorithms (11 points) –** Consider relations Alfa(a, c), Beta(a, b, e), and Gamma(a, d, f) on which we want to perform a natural join (join by equality on the common attribute a). No indexes are available on these relations.

- There are B = 750 buffer pages (main-memory cache)
- Table Alfa has M = 2,500 pages with 80 records per page
- Table Beta has N = 600 pages with 300 records per page
- Table Gamma has O = 1,500 pages with 150 records per page
- The join result of Alfa and Beta has P = 500 pages

What is the **I/O cost** for the following joins algorithms, ignoring the cost of the final writing to disk of join results?

(a) (1 point) Page nested loop join (PNLJ) with Alfa as the outer relation and Beta as the inner relation?

(b) (1 point) Chunk nested loop join (CNLJ) with Gamma as the outer relation and Beta as the inner one?

(c) (1 point) Chunk Nested Loop Join (CNLJ) with Beta as the outer relation and Gamma as the inner one?

In what follows, let us assume that we compute a Sort-Merge Join (SMJ) with Alfa as the outer relation and Gamma as the inner relation.

(d) (1 point). What is the cost of sorting the records in Alfa on attribute a?

(e) (1 point) What is the cost of sorting the records of Gamma on attribute a?

(f) (1 point) What is the cost of the merge phase in the worst-case scenario of the previous SMJ?

(g) (1 point) What is the cost of the merge phase assuming there are no duplicates in the join attribute a?

(h) (1 point) Let us assume we first join Alfa and Beta, and then join the result with Gamma. What is the cost of the final merge phase assuming there are no duplicates in the join attribute a?

Let us consider next a hash-join with Beta as the outer relation and Gamma as the inner one:
(i) (1 point) Assuming no recursive partitioning is needed, detail the cost of the probing (conquer) stage.

(j) (1 point) Assuming no recursive partitioning is needed, detail the cost of the partitioning stage.

(k) (1 point) Assume that the tables do not fit in main memory and that a large number of distinct values will hash to the same bucket using hash function h_p. What should we do in this case?

Q2.2 – **Grace Hash Join (8 points) –** Products has a total of 100 pages and 20 tuples per page. Orders has a total of 50 pages and 50 tuples per page. Assume the hash functions uniformly distribute the data for both tables.

(a) (2 points) If we had 10 buffer pages, how many partitioning phases would we require for Grace Hash Join? Consider which table we should build the hash table in the probing phase on.

(b) (2 points) What is the I/O cost for the Grace Hash Join in that case?

(c) (2 points) For the above question, if we only had 8 buffer pages, how many partitioning phases would be necessary?

(d) (2 points) What will be the I/O cost in that case?

Q2.3 – **Query Optimization – Selectivity Estimation (8 points) -** Consider the following schema:

HORSE(horseId INTEGER PRIMARY KEY, breed VARCHAR(50), age INTEGER);
CAVALIER(cavId INTEGER PRIMARY KEY, name VARCHAR(50), age INTEGER);
RIDE(horseId INTEGER REFERENCES Horse(horseId), cavId INTEGER REFERENCES Cavalier(cavId), date DATE, PRIMARY KEY (horseId, cavId));

We make the following assumptions about the distribution of data:

- $15 \leq$ Cavalier.age < 90
- $5 \leq$ Horse.age < 50
- cavId has 1000 distinct values
- horseId has 1000 distinct values
- breed has 10 distinct values
- we assume attribute independence over all attributes

Consider the following query:

SELECT C.name
FROM Horse H, Cavalier C, Ride R
WHERE H.horseId = R.horseId AND C.cavId = R.cavId
AND (C.age < 30 OR H.breed = "Mustang") AND H.age >= 30;

Compute the selectivity for each of the following predicates from the WHERE clause. Write only your final answer, as a simple fraction.

(a) (2 points) *H.horseId = R.horseId*

(b) (2 points) *C.cavId = R.cavId*

(c) (2 points) H.age >= 30

(d) (2 points) C.age < 30 OR H.breed = "Mustang"

Q2.4 – **Locking, 2PL & strict 2PL (5 points) –** Consider the following schedule, where time flows top to bottom, and Lock means a request for the lock, in either shared (S) or exclusive (X) mode, modifying resources B and F.

T1	T2
Lock_X(B)	
Read(B)	
B := B * 10	
Write(B)	
Lock_X(F)	
Unlock(B)	
	Lock_S(F)
F := B * 100	
Write(F)	
Commit	
Unlock(F)	
	Read(F)
	Unlock(F)
	Lock_S(B)
	Read(B)
	Print(F + B)
	Commit
	Unlock(B)

(a) (1 point) What is printed, assuming we initially have B = 3 and F = 300?

(b) (1 point) Does the execution use 2PL, strict 2PL, or neither?

(c) (1 point) Would moving Unlock(F) in the second transaction to any point after Lock_S(B) make it become 2PL (or keep it 2PL if it was already so) ?

(d) (1 point) Would moving Unlock(F) in the first transaction and Unlock(F) in the second transaction to

the end of their respective transactions change this (or keep it) in strict 2PL?

(e) (1 point) Would moving Unlock(B) in the first transaction and Unlock(F) in the second transaction to the end of their respective transactions change this (or keep it) in strict 2PL?

Q2.5 – **Lock manager (9 points)** – The lock manager answers three kinds of requests:

- request(transaction, resource, mode): this is to allow a transaction to request one lock, in either Shared (S) or eXclusive (X) mode. The manager grants the request if (when) there is no queue and the request is compatible with existing locks. Otherwise, it should queue the request (at the back of the queue) and block the transaction.
- release(transaction, resource): this allows a transaction to release one lock that it holds.
- upgrade(transaction, resource): this allows a transaction to upgrade a held lock from Shared to Exclusive. Such a request has priority over any existing queued requests (it is placed in the front of the queue if it cannot proceed).

In this context, we will have 3 transactions (T1, T2, T3) and 2 resources that they will try to lock (A and B). After each lock request is processed, fill out the following two tables that store the information for what locks are held:

The following lock requests are made in this order:

a) (1 point) request(T1, A, X)

Transaction	**Locks held**
T_1	
T_2	
T_3	

Resource	**Locks**	**Queue**
A		
B		

b) (1 point) request(T2, B, S)

Transaction	Locks held
T_1	
T_2	
T_3	

Resource	Locks	Queue
A		
B		

c) (1 point) request(T3, B, X)

Transaction	Locks held
T_1	
T_2	
T_3	

Resource	Locks	Queue
A		
B		

d) (1 point) request(T3, A, S)

Transaction	Locks held
T_1	
T_2	
T_3	

Resource	Locks	Queue
A		
B		

e) (1 point) release(T1, A)

Transaction	Locks held
T_1	
T_2	
T_3	

Resource	Locks	Queue
A		

B		

f) (1 point) request(T2, A, S)

Transaction	Locks held
T_1	
T_2	
T_3	

Resource	Locks	Queue
A		
B		

g) (1 point) request(T1, B, S)

Transaction	Locks held
T_1	
T_2	
T_3	

Resource	Locks	Queue
A		
B		

h) (1 point) upgrade(T2, B, X)

Transaction	Locks held
T_1	
T_2	
T_3	

Resource	Locks	Queue
A		
B		

i) (1 point) release(T1, B)

Transaction	Locks held
T_1	
T_2	
T_3	

Resource	Locks	Queue
A		
B		

Q2.6 - **Concurrency control (8 points)** Consider the following schedule (time flows top-to-bottom, each line corresponds to one clock "tick"). R stands for Read, W stands for Write, Com stands for Commit.

	T1	T2	T3	T4
1	R(A)			
2		R(A)		
3			R(C)	
4			W(C)	
5		R(B)		
6		W(B)		
7	R(B)			
8				R(B)
9	R(C)			
10				R(C)
11				W(B)
12		COM		
13			COM	
14	COM			
15				COM

(a) (1 point) Is this schedule serial?

(b) (2 points) Draw the conflict graph for this schedule.

(c) (1 point) Is this schedule conflict serializable?

(d) (2 points) Which of the two locking protocols learned in class could have produced this schedule?

(e) (2 points) Which of the following schedules below are conflict equivalent to the schedule above (cross the ones that are not conflict equivalent)?

A. T3, T1, T2, T4

B. T2, T3, T1, T4

C. T4, T3, T1, T2

D. T3, T2, T1, T4

Q2.7 – **PDBMS (12 points)** Assume that you have access to the following two relations:

```
Dungeon(dungeonId, name, location, surface, owner)
Dragon(dragonId, name, type, weight)
```

The main parameters for this question are summarized in the table below:

Parameter	Symbol	value
Number of nodes	M	4
Size of page (KB)	s	4
Pages in RAM per node	B	8
Size of Dungeon	$[D_g]$	128
Size of Dragon	$[D_r]$	4
Time for each I/O (ms)	t	5

Assume we have 4 nodes, each with 8 pages in memory for joins. We want to measure time, assuming that an I/O takes 5ms. For the following questions, when we ask for execution time, we are only concerned with the time associated with I/Os (e.g., CPU and other costs are negligible). Assume that we can send individual records over the network with no overhead in terms of network cost.
The first two questions will deal with the following query:

```
SELECT DG.name, DR.name, DG.surface, DR.type
FROM Dungeon DG, Dragon DR
```

```
WHERE DG.owner = DR.name;
```

(a) (2 points) Assuming the Dungeon and Dragon relations are both round-robin partitioned across the 4 nodes, what is the largest amount of data we ship across the network, in KB, of the query above, assuming we want to execute a Sort Merge Join?

(b) (2 points) Under the same assumption as before, what would be the expected amount of data we ship across the network, assuming uniform distribution of data by the round-robin partition with respect to the domain of the owner / name attributes.

(c) (2 points) Assuming the Dungeon relation starts out hash-partitioned on the "location" key across the 4 machines, and that 75% of Dungeon gets mapped to one machine, how long will it take to complete a parallel scan of Dungeon?

Suppose we now add another table, with the following schema:

Rider(riderId, name, dragonType, salary)

which has 40,000 pages and is round-robin partitioned across 4 new machines with only this data.

The next two questions deal with the following query:

```
SELECT DR.name, R.name
FROM Dragon DR, Rider R
WHERE R.dragonType = DR.type;
```

(d) (2 point) What is the name of the join strategy that provides the lowest possible network cost (amount of data shipped) to execute the query above? Explain why others would not be a good idea.

(e) (2 points) What is the amount of data shipped in KB to execute that join strategy?

(f) (2 points) We wish to compute the square root of the sum of the squared salary of each rider, as in:

SELECT SQRT(SUM(salary * salary)) FROM Rider

Do we need to repartition the data? What would be a global and local aggregate function that would allow us to efficiently compute the result via hierarchical aggregation?

Q2.8 – **Query optimization – System R / Selinger Algorithm (15 points)** For the following questions, assume the following:

- The assumptions about uniformity with attribute domains and independence between attributes.
- We use System R defaults whenever selectivity estimation is not possible.
- Primary key IDs are sequential, starting from 1.
- To simplify, our optimizer does not consider interesting orders.

CREATE TABLE Engineer (eid INTEGER PRIMARY KEY, name VARCHAR(32), field VARCHAR(64)), years_experience INTEGER) ;

- Table Engineer has 25,000 records in 500 pages.
- Index 1: Clustered(field). There are 130 unique fields.
- Index 2: Unclustered(years_experience) and there are 11 unique values in the range [0,10] for this attribute.

CREATE TABLE Application (eid INTEGER REFERENCES Engineer, cid INTEGER REFERENCES Company, status TEXT, referral INTEGER, (eid, cid) PRIMARY KEY)

- Table Application has 100,000 records in 10,000 pages.
- Index 3: Clustered(cid, eid).
- Status has 10 unique values.

CREATE TABLE Company (cid INTEGER PRIMARY KEY, open_roles INTEGER)

- Table Company has 500 records in 100 pages.
- Index 4 : Unclustered(cid).
- Index 5: Clustered(open_roles) and there are 500 unique values in the range [1,500] for this attribute.

We consider the following query:

```
SELECT Engineer.name, Company.open_roles, Application.referral
FROM Engineer, Application, Company
WHERE Engineer.eid = Application.eid.                    -- selectivity 1
AND Application.cid = Company.cid                        -- selectivity 2
AND Engineer.years_experience > 6                        -- selectivity 3
AND (Engineer.field='Naval' OR Company.open_roles <= 50) -- selectivity 4
AND NOT Application.status = 'Pending'                   -- selectivity 5
ORDER BY Company.open_roles;
```

Write the reduction factor (RF) for each clause.

(a) (1 point) *Engineer.eid = Application.eid*

(b) (1point) *Application.cid = Company.cid*

(c) (1 point) *Engineer.years_experience > 6*

(d) (2 points) *(Engineer.field='Naval' OR Company.open_roles <= 50)*

(e) (1 point) *NOT Application.status = 'pending'*

(f) (3 points) For each predicate, mark the first pass of Selinger's algorithm that uses its selectivity to estimate output size.

i. Selectivity 1: Pass 1, Pass 2, or Pass 3.
ii. Selectivity 2: Pass 1, Pass 2, or Pass 3.
iii. Selectivity 3: Pass 1, Pass 2, or Pass 3.
iv. Selectivity 4: Pass 1, Pass 2, or Pass 3.
v. Selectivity 5: Pass 1, Pass 2, or Pass 3.

(g) (1.5 points) Mark the choices for all access plans that would be considered in pass 2 of the Selinger algorithm.

i. Engineer JOIN Application (800 IOs)
ii. Application JOIN Engineer (750 IOs)
iii. Engineer JOIN Company (470 IOs)
iv. Company JOIN Engineer (525 IOs)
v. Application JOIN Company (600 IOs)
vi. Company JOIN Application (575 IOs)

(h) (1.5 points) Mark the choices from the previous question for all access plans that would be chosen at the end of pass 2 of the Selinger algorithm.

(i) (1.5 points) Mark the choices for all plans that would be considered in pass 3.

i. Company JOIN (Application JOIN Engineer) (175,000 IOs)
ii. Company JOIN (Engineer JOIN Application) (150,000 IOs)
iii. Application JOIN (Company JOIN Engineer) (155,000 IOs)
iv. Application JOIN (Company JOIN Engineer) (160,000 IOs)
v. Engineer JOIN (Company JOIN Application) (215,000 IOs)
vi. (Company JOIN Application) JOIN Engineer(180,000 IOs)
vii. (Application JOIN Company) JOIN Engineer (200,000 IOs)
viii. (Application JOIN Engineer) JOIN Company (194,000 IOs)
ix. (Engineer JOIN Application) JOIN Company (195,000 IOs)
x. (Engineer JOIN Company) JOIN Application (165,000 IOs)

(j) (1.5 points) Mark the choices from the previous question for all plans that would be chosen at the end of pass3.

END OF PAPER